👑 CODERKING STUDIO

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EMBEDDED SYSTEMS & STORAGE

SQLite: 400 Crore Devices on a Single C File

Deep dive into sqlite3.c: zero network lag, in-process memory calls, and military-grade branch test coverage.

💾 1. Zero-Network Architecture

Unlike Postgres or MySQL, SQLite is not a server process. It compiles directly into your host binary.

- **Direct Function Calls:** Eliminates TCP/socket latency completely.
- **In-Memory & File:** Single disk file storage with full ACID transactions.

📦 2. The 8MB sqlite3.c Amalgamation

The entire database engine is bundled into a single C file:

```
// Direct embedding in C/C++
#include "sqlite3.h"
sqlite3 *db;
sqlite3_open(":memory:", &db);
```

Allows compilers to inline functions across the whole engine.

🌲 3. B-Tree & WAL Mode Internals

High-concurrency secrets of SQLite:

- **Write-Ahead Logging (WAL):** Readers never block writers, writers never block readers.
- **B-Tree Paging:** 4KB disk page cache with atomic commits.

🛡️ 4. 100% Branch Coverage Testing

The aviation standard of software reliability:

```
// SQLite has 600x more test code
// than actual library implementation co
de!
```

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Official GitHub: **github.com/shubh74/coderking-roadmaps**